

FSD (Full Stack Development)

Exam-Ready Notes

1. Difference Between SQL and NoSQL

SQL and NoSQL are two types of databases. SQL is mainly used for structured data, while NoSQL is used for flexible and large amounts of data.

SQL	NoSQL
SQL means Structured Query Language.	NoSQL means Not Only SQL.
Data is stored in tables.	Data can be stored as documents, key-value, graphs, etc.
It has a fixed schema.	It has a flexible schema.
Uses rows and columns.	Uses different data formats.
Supports ACID transactions.	Usually focuses on availability and scalability.
Suitable for structured data.	Suitable for unstructured or semi-structured data.
Supports JOIN operations.	Usually avoids complex JOINS.
Mainly uses vertical scaling.	Mainly uses horizontal scaling.
Used in banking systems.	Used in social media and big-data applications.
Uses standard SQL queries.	Query syntax depends on the database.
Examples: MySQL, Oracle, PostgreSQL.	Examples: MongoDB, Cassandra, Redis.

Conclusion: SQL is suitable for structured and highly consistent data. NoSQL is suitable for flexible, large-scale, and changing data.

2. Working of API and Endpoints

What is an API?

API stands for Application Programming Interface. An API allows two applications to communicate with each other. It receives a request, processes it, and sends a response.

Example: A weather application uses an API to get weather information from a server.

Working of API

1. User sends a request.
2. The request reaches the API.
3. API checks and processes the request.
4. Server performs the required operation.
5. Database provides the required data.
6. API sends the response to the user.

Flow: User → Client → API → Server → Database → API → User

What is an Endpoint?

An endpoint is a specific URL used to perform an operation through an API.

Method	Example	Purpose
GET	/users	Get all users
GET	/users/1	Get one user
POST	/users	Add a new user
PUT	/users/1	Update a user
DELETE	/users/1	Delete a user

Conclusion: API helps applications communicate, while an endpoint specifies where a particular API operation is performed.

3. Spring Architecture and Spring Boot

What is Spring?

Spring is a Java framework used to develop web and enterprise applications. It divides an application into different layers. Each layer performs a specific task.

Spring Architecture

1. Presentation Layer

- Interacts with the user.
- Uses controllers.
- Receives user requests.

2. Business Layer

- Contains business logic.
- Uses service classes.
- Performs required operations.

3. Data Access Layer

- Connects the application with the database.
- Uses repositories or DAO classes.

4. Database Layer

- Stores application data.
- Example: MySQL.

Main Spring Modules

- Core Container
- AOP
- Data Access
- Web
- Security
- Testing

What is Spring Boot?

Spring Boot is a framework built on Spring. It makes Spring application development easier and faster by reducing configuration.

Features of Spring Boot

1. Auto Configuration
2. Embedded Tomcat Server
3. Starter Dependencies
4. Less configuration
5. Faster development

Working Flow: Client □ Controller □ Service □ Repository □ Database

Spring vs Spring Boot

Spring	Spring Boot
Requires more configuration.	Requires less configuration.
Usually needs an external server setup.	Provides an embedded server.
More setup is required.	Easy and quick setup.
Development can require more configuration.	Makes development faster.

Conclusion: Spring provides the framework, while Spring Boot makes Spring application development simpler and faster.

4. JUnit

What is JUnit?

JUnit is a Java testing framework used to test Java programs automatically. It helps developers check whether the program is working correctly.

Importance of JUnit

- Finds errors early.
- Saves testing time.
- Improves code quality.
- Supports automatic testing.
- Makes applications more reliable.

Common JUnit Annotations

Annotation	Purpose
@Test	Marks a method as a test method.
@BeforeEach	Runs before every test.
@AfterEach	Runs after every test.
@BeforeAll	Runs once before all tests.
@AfterAll	Runs once after all tests.

Advantages of JUnit

1. Easy to use.
2. Finds errors quickly.
3. Supports repeated testing.
4. Saves time.
5. Improves software reliability.

Conclusion: JUnit helps developers test Java programs automatically and find errors before the application is released.

5. Spring Data JPA and ORM

What is ORM?

ORM stands for Object Relational Mapping. ORM is a technique that connects Java objects with database tables. It allows developers to work with Java objects instead of writing many SQL queries.

Example

ID	Name
1	Ravi

A Java object containing id = 1 and name = Ravi can be mapped to this database record by ORM.

Benefits of ORM

1. Reduces SQL code.
2. Makes development faster.
3. Makes code easier to maintain.
4. Improves readability.
5. Connects objects with database tables.

What is Spring Data JPA?

Spring Data JPA is a Spring module used to simplify database operations. It provides ready-made methods for common CRUD operations.

CRUD means: Create, Read, Update, Delete

Architecture Flow: Controller ☐ Service ☐ Repository ☐ Hibernate ☐ Database

Common JPA Methods

Method	Purpose
save()	Insert or update data
findAll()	Get all records
findById()	Get one record
deleteById()	Delete a record
count()	Count records

JPA vs Hibernate

JPA	Hibernate
JPA is a specification.	Hibernate is an implementation of JPA.
It defines rules and interfaces.	It provides the actual functionality.
It is portable.	It provides additional features.

Advantages of Spring Data JPA

1. Less code is required.
2. Provides automatic CRUD operations.
3. Makes database connection easier.
4. Supports sorting and pagination.

5. Improves development speed.

Conclusion: ORM maps Java objects to database tables, while Spring Data JPA makes database operations easier using JPA.

Quick Revision

Topic	Simple Meaning
SQL	Stores structured data
NoSQL	Stores flexible data
API	Helps applications communicate
Endpoint	URL used for an API operation
Spring	Java application framework
Spring Boot	Makes Spring development easier
JUnit	Java testing framework
ORM	Maps Java objects to database tables
Spring Data JPA	Simplifies database operations